Classical ML — Final Project

Portfolio Level: 3 ta Kaggle Dataset ustida to'liq ML Pipeline

EDA → Preprocessing → Feature Engineering → Feature Importance → Modeling → Tuning → Comparison

XGBoost | LightGBM | CatBoost

Agenda

- Loyiha maqsadi va metodologiya
- Dataset 1 — Credit Card Customers (Customer Churn)
- Dataset 2 — Adult Census Income
- Dataset 3 — Bank Customer Churn Prediction
- 3 ta dataset bo'yicha umumiy taqqoslash va xulosalar
- Tavsiyalar

Loyiha maqsadi va metodologiya

Har bir talaba 3 ta real-world Kaggle dataset ustida to'liq Machine Learning pipeline yaratishi kerak.

EDA → Data Cleaning → Feature Engineering → Feature Importance → Model Training → Hyperparameter Tuning → Model Comparison

Modellar: XGBoost, LightGBM, CatBoost | Tuning: RandomizedSearchCV | Metrikalar: Accuracy, Precision, Recall, F1, ROC-AUC

3 ta Dataset — umumiy ko'rinish

#	Dataset	Manba (Kaggle)	Qatorlar	Target	Balans
1	Credit Card Customers	sakshigoyal7/credit-card-customers	10,127	Attrition_Flag	83.9% / 16.1%
2	Adult Census Income	uciml/adult-census-income	32,561	income	75.2% / 24.8%
3	Bank Customer Churn	shrutimechlearn/churn-modelling	10,000	Exited	79.6% / 20.4%

Barcha 3 dataset ham imbalanced binary classification muammosi

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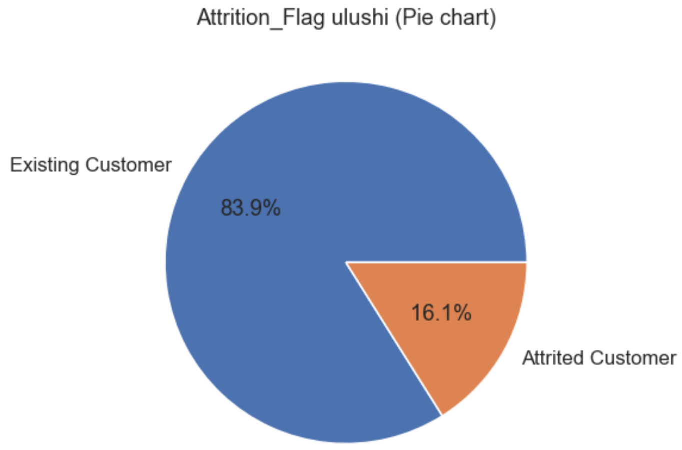
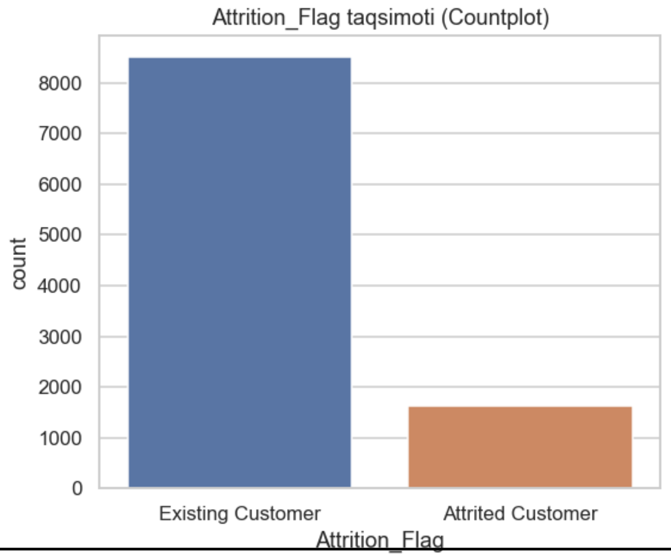
Credit Card Customers

Customer Churn Prediction — 10,127 mijoz, 20 feature

Dataset 1 — EDA asosiy topilmalar

10,127 / 20	**0 / 0**	**83.9% / 16.1%**	**Total_Trans_Ct**
Qatorlar / Ustunlar	Missing / Duplicate	Target balans	Eng kuchli signal

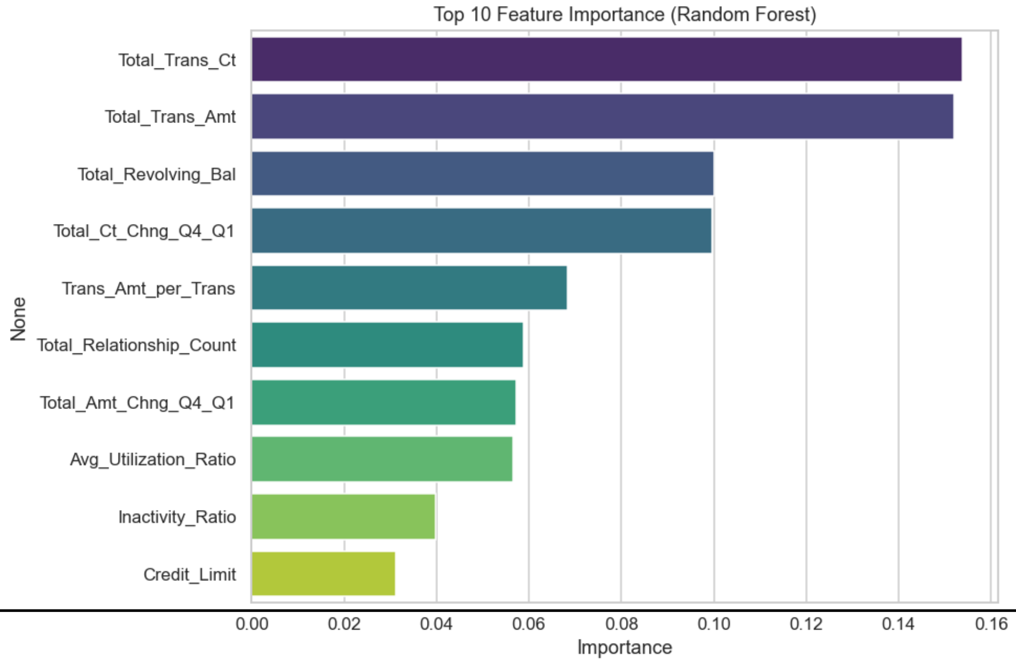




Dataset 1 — Feature Engineering & Feature Importance

Yangi featurelar:

- Trans_Amt_per_Trans = Total_Trans_Amt / Total_Trans_Ct
- Inactivity_Ratio = Months_Inactive_12_mon / Months_on_book



Dataset 1 — Model Comparison (tuned)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
XGBoost	0.9719	0.9497	0.8708	0.9085	0.9924
LightGBM	0.9704	0.9316	0.8800	0.9051	0.9923
CatBoost	0.9719	0.9497	0.8708	0.9085	0.9931

Eng yaxshi model: CatBoost — eng yuqori ROC-AUC (0.9931), production uchun tavsiya etiladi

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Adult Census Income

32,561 qator, 14 ustun — daromad bashorati

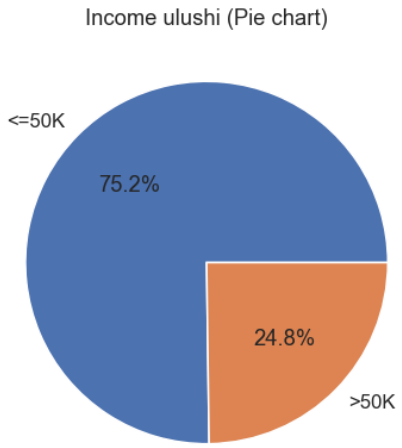
Dataset 2 — EDA asosiy topilmalar

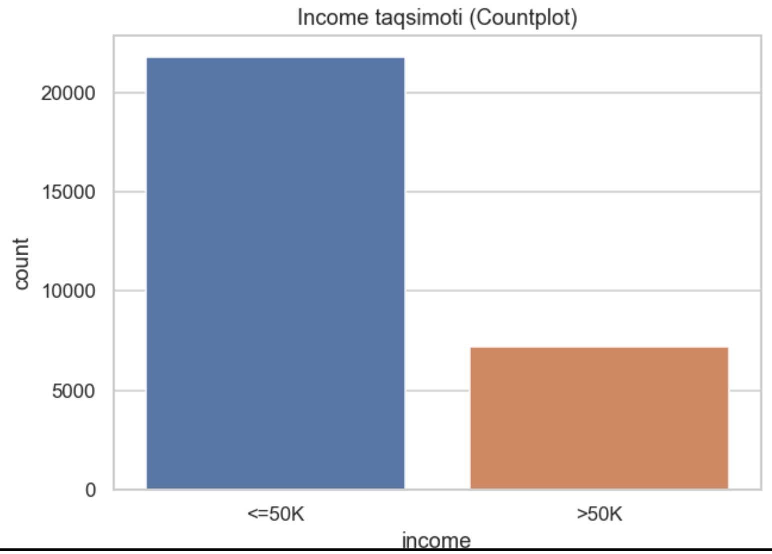
32,561 / 14
Qatorlar / Ustunlar

4,262 / 3,591
Missing / Duplicate

75.2% / 24.8%
Target balans

age
Eng kuchli feature

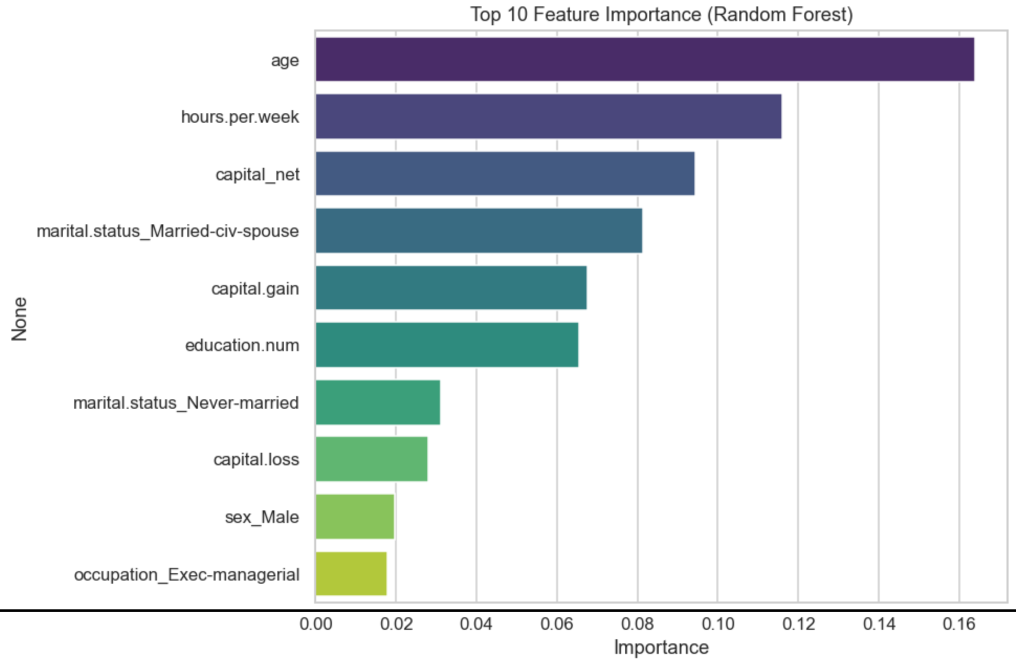




Dataset 2 — Feature Engineering & Feature Importance

Yangi featurelar:

- capital_net = capital.gain – capital.loss
- age_group = yosh 6 ta guruhga bo'lingan (<25 ... 65+)



Dataset 2 — Model Comparison (tuned)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
XGBoost	0.8693	0.7822	0.6553	0.7131	0.9292
LightGBM	0.8697	0.7830	0.6560	0.7139	0.9279
CatBoost	0.8688	0.7845	0.6490	0.7104	0.9285

Eng yaxshi model: LightGBM — eng yuqori F1/Accuracy va eng tez train vaqti

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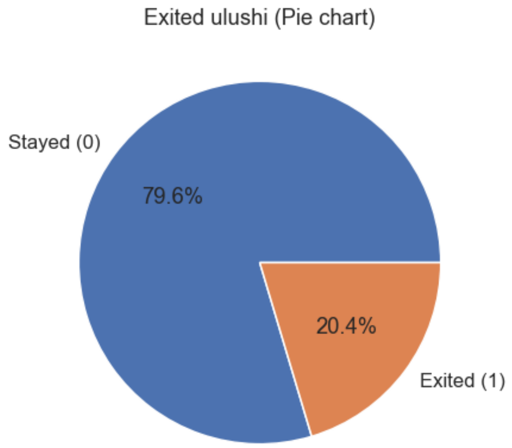
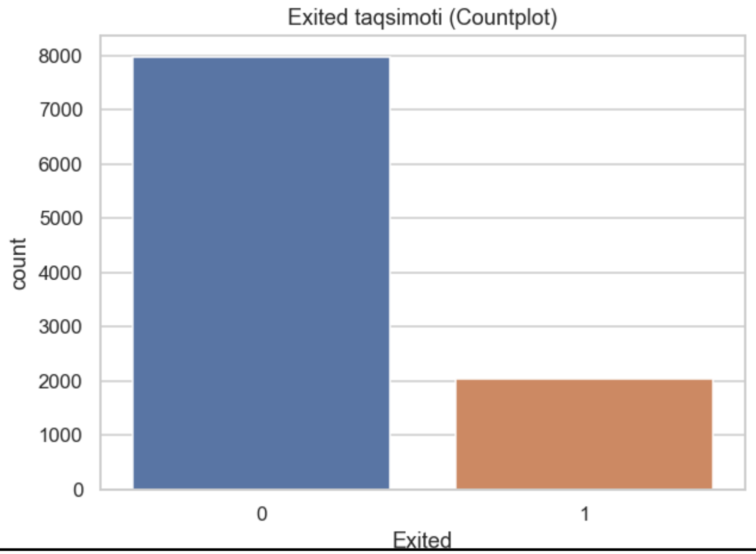
Bank Customer Churn

10,000 mijoz, 11 ustun — bank mijozlari churn'i

Dataset 3 — EDA asosiy topilmalar

10,000 / 11	**0 / 0**	**79.6% / 20.4%**	**Age**
Qatorlar / Ustunlar	Missing / Duplicate	Target balans	Eng kuchli feature

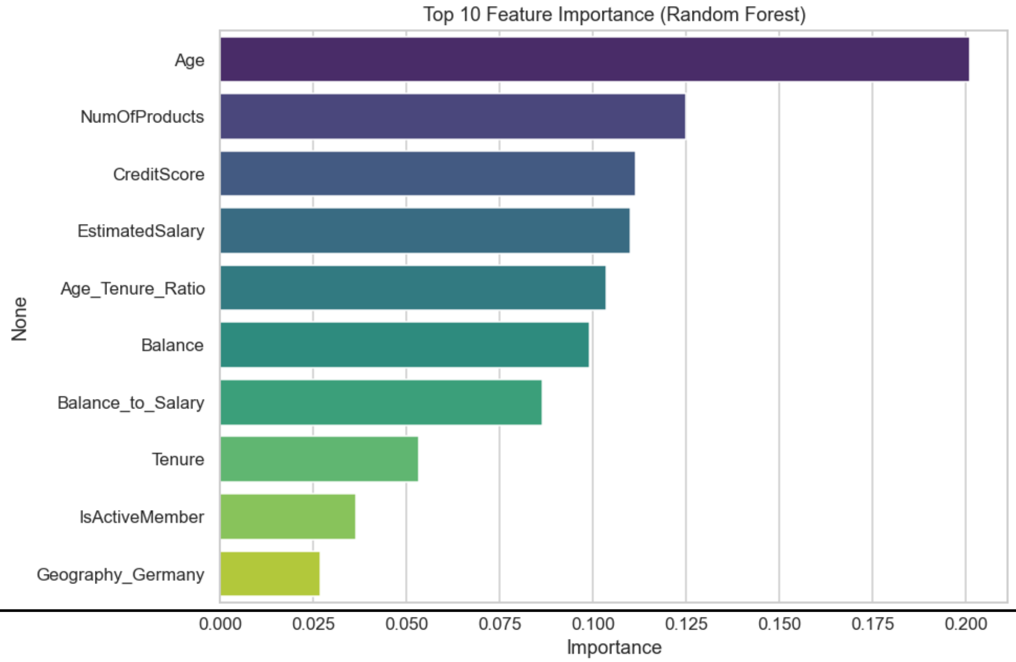




Dataset 3 — Feature Engineering & Feature Importance

Yangi featurelar:

- Balance_to_Salary = Balance / EstimatedSalary
- Age_Tenure_Ratio = Age / Tenure



Dataset 3 — Model Comparison (tuned)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
XGBoost	0.8610	0.7529	0.4717	0.5801	0.8460
LightGBM	0.8690	0.7843	0.4914	0.6042	0.8661
CatBoost	0.8665	0.7713	0.4889	0.5985	0.8633

Eng yaxshi model: LightGBM — barcha metrikalarda yetakchi (Recall umumiy past, class imbalance sababli)

Barcha datasetlar bo'yicha eng yaxshi modellar

Dataset	Eng yaxshi model	F1	ROC-AUC	Asosiy sabab
1. Credit Card Customers	CatBoost	0.9085	0.9931	Eng yuqori ROC-AUC
2. Adult Census Income	LightGBM	0.7139	0.9279	Eng yuqori F1, eng tez train
3. Bank Customer Churn	LightGBM	0.6042	0.8661	Barcha metrikalarda yetakchi

Umumiy xulosalar

- Har uchala datasetda ham target imbalanced — Accuracy emas, F1/ROC-AUC asosiy metrika bo'lishi kerak
- Behavioral/transactional featurelar (masalan Total_Trans_Ct, IsActiveMember) demografik featurelardan kuchliroq signal beradi
- Qo'lda yaratilgan yangi featurelar (feature engineering) har uchala datasetda ham top-10 muhim feature qatoriga kirdi
- Hyperparameter tuning natijalarni kichik miqdorda yaxshiladi (F1 +0.01-0.03) — default sozlamalar allaqachon kuchli baseline
- XGBoost, LightGBM, CatBoost deyarli teng natija ko'rsatadi — tanlov ko'pincha tezlik va operatsion talablarga bog'liq

Rahmat!

To'liq kod, notebooklar va hisobotlar GitHub repositoriyda

Classical-ML-Final-Project/README.md